1 FIFO vs Pro Rata Order Allocation

FIFO vs Pro Rata order allocation methods dictate participant behavior. Both methods have advantages and drawbacks:

(a) FIFO fills orders (at the best price level) based on which orders arrived first.

1. FIFO rewards speed, which encourages rapid price discovery. In order to guarantee your order is filled, you have to post your price first. This gives tighter spreads.
2. FIFO is the lowest latency algorithm. It only requires a look at the head of the PriceLevel linked list. It fills the head, deletes if empty, and moves on (so $O(1)$ for each fill).

(b) Pro Rata fills orders based on the proportion of total volume that each order holds.

1. Every order at a given price level must be iterated through to calculate the proportions (so $O(n)$ for each fill).
2. Quote stuffing may become a problem. Traders may also be incentivized to post larger orders than they want to fill, with hopes of capturing their desired number of orders.
3. Allows for greater competition (not just HFTs).
4. Pro Rata implementation also deepens the book - by incentivizing large orders, huge amounts of stock can be dumped without changing the price, as there's enough resting volume to soak it up.

2 Hybrid Allocation Algorithm Proposal

Pure FIFO gives no depth. Pure Pro Rata gives no price finding. This encourages us to find some mixture of both, allowing us to capture the advantages of each. The easiest way to do this would be to allocate some proportion of shares Pro Rata, and the rest FIFO. But this fails to capture the fact that FIFO is better under some conditions, and worse under others. This brings us to my proposal.

When we have a deep, stable, book, FIFO is preferred. It's the faster algorithm, so when we don't need to guard against massive orders drastically changing the price, it is a superior allocation method to Pro Rata.

When we have a shallow, unstable, book, some amount of Pro Rata is required to incentivize liquidity providers (LPs) to post large orders (especially given the large spreads present in shallow, unstable, books). However, some amount of FIFO is always required to incentivize HFT participation and thus rapid price finding.

We begin by introducing a few variables:

(a) Effective liquidity (L_{eff}): a measure of market depth.

At the best price, let V = total shares and N = total orders. Now write

$$L_{eff} = V\sqrt{N}$$

So as volume increases, L_{eff} increases proportionally. And as number of orders increases, L_{eff} has a diminishing increase.

This represents the most "aggressive" approach we can take. By only considering the best price level, we are introducing infinite decay into our L_{eff} function. This isn't entirely accurate - a book with one share at the best price and a thousand shares at a price one tick worse is far more stable than a book possessing only the single share at the best price. So we introduce an exponential decay into our liquidity function:

Given best price P and current price P_i (with associated share/order counts of V_i/N_i), L_{eff} can be written as

$$L_{eff}(P_i) = (V_i\sqrt{N_i}) \times \begin{cases} 1.0 & \text{if } P_i = P \\ 0.5 & \text{if } 0 < |P_i - P| \leq 0.10P \\ 0.25 & \text{if } 0.10P < |P_i - P| \leq 0.20P \\ 0 & \text{else} \end{cases}$$

We now write total L_{eff} as

$$L_{eff}(P) = \sum_{i \in \mathbb{N}} L_{eff}(P_i)$$

This can be better implemented as

$$L_{eff}(P) = \sum_{p \in \Omega} L_{eff}(p)$$

for $\Omega = \{0.8P, ..., 1.2P\}$.

Of course, this is far from perfect. Theoretically, resting volume at every price level should contribute at least marginally to the liquidity of the book. However, this would be extremely computationally intensive. This idea could be further simplified by only considering resting volume at price levels directly adjacent to the best price (i.e. by removing the 0.25 decay level).

****It may be valuable to test the allocation algorithm given various decay approaches to gauge performance impact.****

(b) Heat (H_c and H_p): a measure of volatility.

We introduce two "heat" functions, one for cancel volatility and one for price volatility.

1. Cancel volatility spikes when orders are canceled, and decays at a rate d_c. decay can be applied either at a constant time interval, or whenever an order is filled.

   ****It may be valuable to test different decay approaches to see how they affect the effectiveness of the algorithm.****

   We write cancel volatility H_c as:

   - Every cancel: $H_c \mathrel{+}= 1$
   - Every order fill: $H_c \mathrel{*}= d_c$

2. Price volatility spikes when the price of a security (midpoint between best bid and best ask) changes. This is a point of danger as a single-share order posted at an abysmal price could massively impact H_p. We can guard against this by only updating price on a fixed time interval (as opposed to continuously checking). We could also mandate that a certain volume be present at a new "best price" for us to confidently conclude that the price has changed. Price volatility H_p is incremented with Δ price any time it changes, and decays on a fixed time interval at rate d_p.

****It may be valuable to test different time increments for d_p application to see how they affect the effectiveness of the algorithm.****

We write price volatility H_p as:

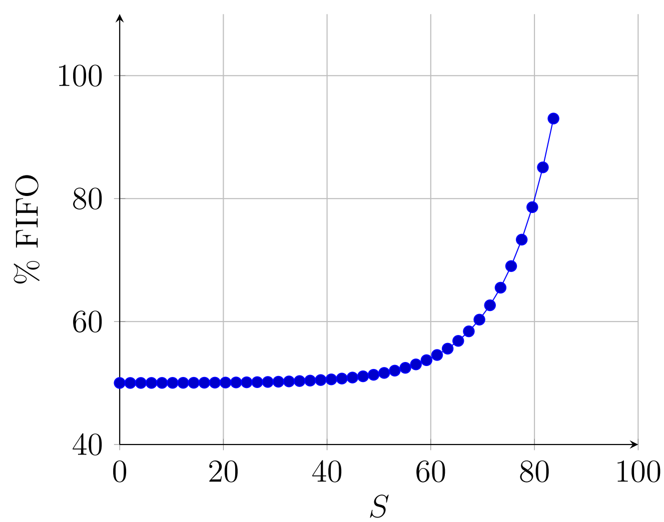
- Every price change: $H_p += \Delta$ price
- Every 10ms: $H_p *= 0.95$

Using $L_{eff}, H_c,$ and H_p, we can construct a holistic measure of book stability S:

$$S = \frac{L_{eff}}{H_c + H_p + 1}$$

High stability demands high L_{eff} and low total heat. But very low L_{eff} or very high total heat will constitute low stability.

Now we aim to map S to some percentage FIFO allocation to use in our matching engine logic. I propose allocating a minimum fixed rate of 50% FIFO allocation. The remaining 50% should be allocated according to S. High S should return a higher FIFO rate, and $S = 0$ should map to entirely Pro Rata. We have several options for a curve to connect these two extremes, but I propose an exponential model. This will keep the rate of FIFO % increase small in periods of low stability, and once the model is "sure" that the books are stable, it increases the FIFO allocation % rapidly.



This plot isn't meant to represent the actual function mapping $S \to \%\ FIFO$; rather, it should provide an example for the shape of the curve. A specific curve can be fit once sample S values are taken in a simulated market.

****We first need to run a simulation with pure FIFO allocation and keep track of S - we can use the bounds we find to write an accurate function.****